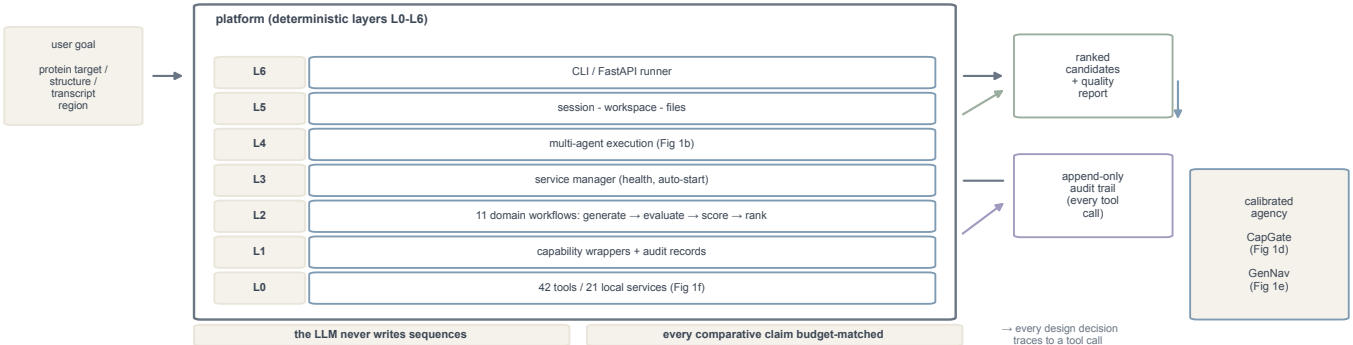
RNADesignAgent: a calibrated, tool-first agent for RNA design
user goal
protein target / structure / transcript region
platform (deterministic layers L0-L6)
L6
CLI / FastAPI runner
L5
session - workspace - files
L4
multi-agent execution (Fig 1b)
L3
service manager (health, auto-start)
L2
11 domain workflows: generate → evaluate → score → rank
L1
capability wrappers + audit records
L0
42 tools / 21 local services (Fig 1f)
ranked candidates + quality report
append-only audit trail (every tool call)
calibrated agency
CapGate (Fig 1d)
GenNav (Fig 1e)
the LLM never writes sequences
every comparative claim budget-matched
→ every design decision traces to a tool call